



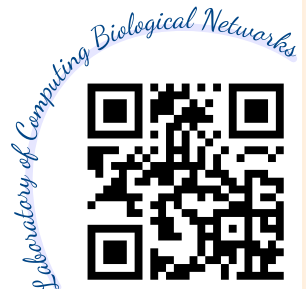
Lecture 14

2025-12-03

Continuously labeled network; Modeling network functions; Simple and complex cells; Hubel–Wiesel model

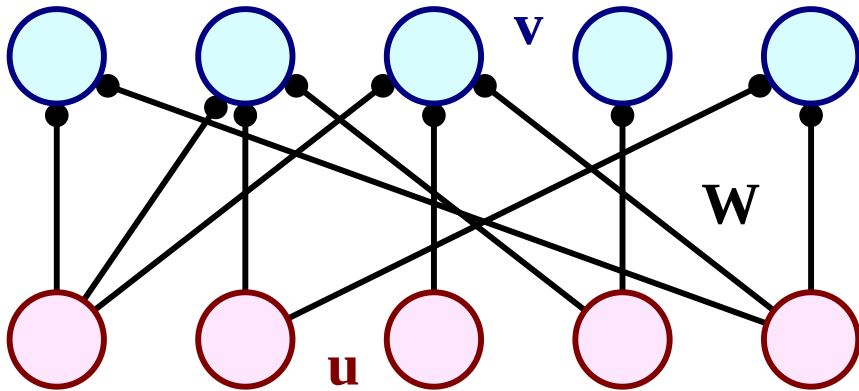
- Online slides: [lec14-functional networks.html](https://lec14-functional-networks.html)
- Code: [code14.ipynb](#)
- Homework: [hw14.pdf](#)

Username: **cns**, Password: **nycu2025**

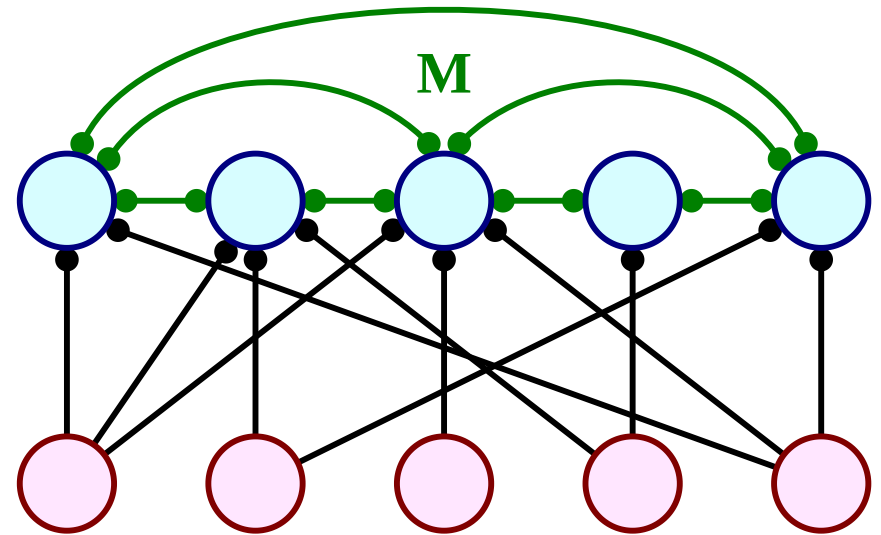


Discrete networks (from last lecture)

Feedforward



Recurrent



$$\begin{aligned}\tau_r \frac{dv}{dt} &= -\mathbf{v} + \mathbf{F}(\mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}) \\ &= -\mathbf{v} + \mathbf{F}(\mathbf{h} + \mathbf{M} \cdot \mathbf{v})\end{aligned}$$

Continuously labeled network

Discrete formulation of a recurrent neural network dynamics

$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{F}(\mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}) = -\mathbf{v} + \mathbf{F}(\mathbf{h} + \mathbf{M} \cdot \mathbf{v})$$

→ Neurons are indexed by discrete variables, e.g., a , b , or a' in $W_{a,b}$ or $M_{a,a'}$.

Consider densely distributed neurons across some aspect of their selectivity, e.g., preferred orientation angles, phases, etc... We can instead label the neurons with continuous variables $a, b \rightarrow \theta_a, \theta_b$ so that $W_{a,b} = W(\theta_a, \theta_b)$ and $M_{a,a'} = M(\theta_a, \theta_{a'})$.

Some simplifying constraints that will be considered:

- Stimuli characterized by a single angle Θ
- Rotational invariance: $W_{a,b} = W(\theta_a - \theta_b)$, $M_{a,a'} = M(\theta_a - \theta_{a'})$
- Uniform coverage: neuronal density ρ_θ is constant

Continuous formulation of network dynamics

– similar to so-called neural field models

Approximate the sums by integrals over θ' multiplied by the density factor ρ_θ :

$$\begin{aligned}\tau_r \frac{dv(\theta)}{dt} &= -v(\theta) + F \left(\rho_\theta \int_{-\pi}^{\pi} d\theta' [W(\theta, \theta')u(\theta') + M(\theta, \theta')v(\theta')] \right) \\ &= -v(\theta) + F \left(h(\theta) + \rho_\theta \int_{-\pi}^{\pi} d\theta' M(\theta, \theta')v(\theta') \right)\end{aligned}$$

Feed forward network

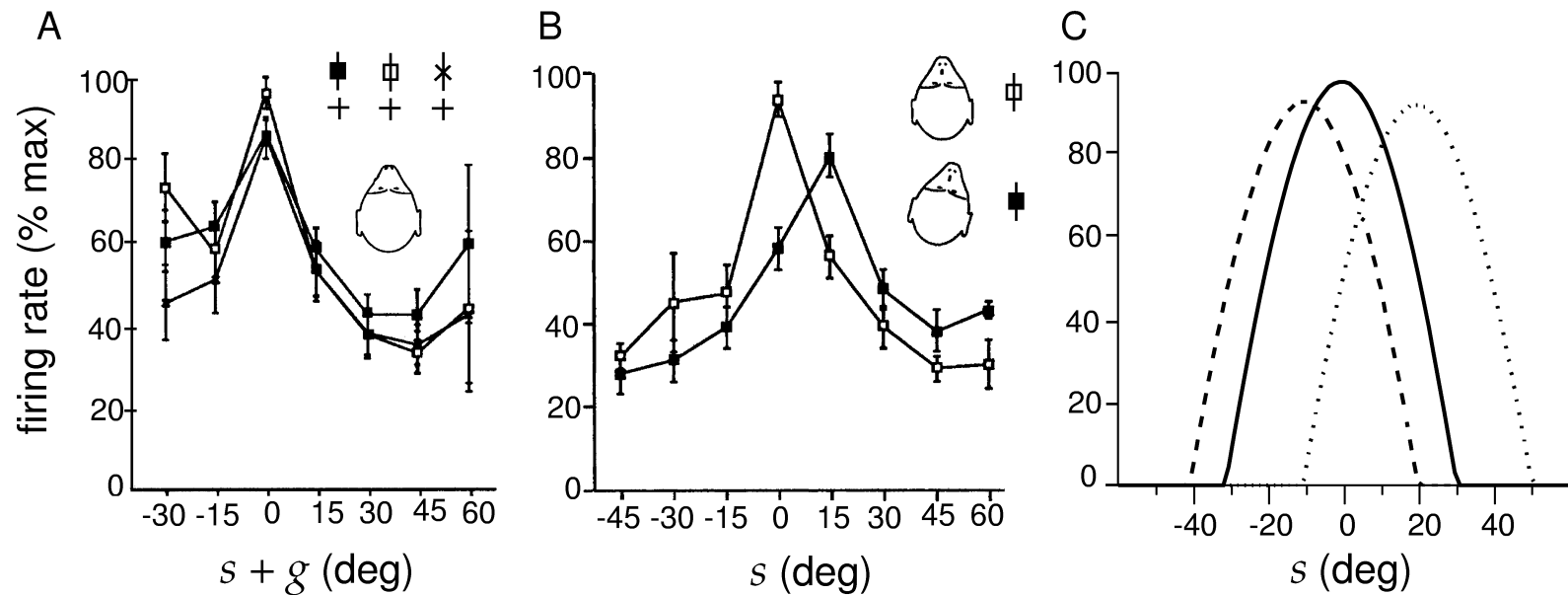
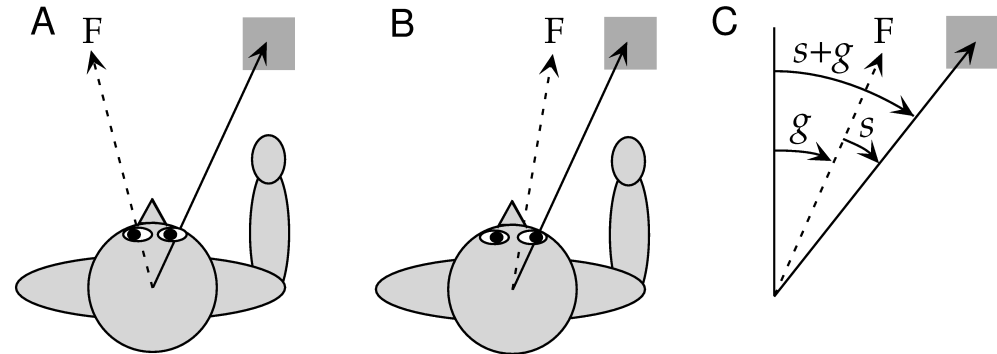
$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + F \left(\rho_\theta \int_{-\pi}^{\pi} d\theta' W(\theta, \theta')u(\theta') \right)$$

At steady state: $v_\infty = F \left(\rho_\theta \int_{-\pi}^{\pi} d\theta' W(\theta, \theta')u(\theta') \right)$

Example: Neural coordinate transformations

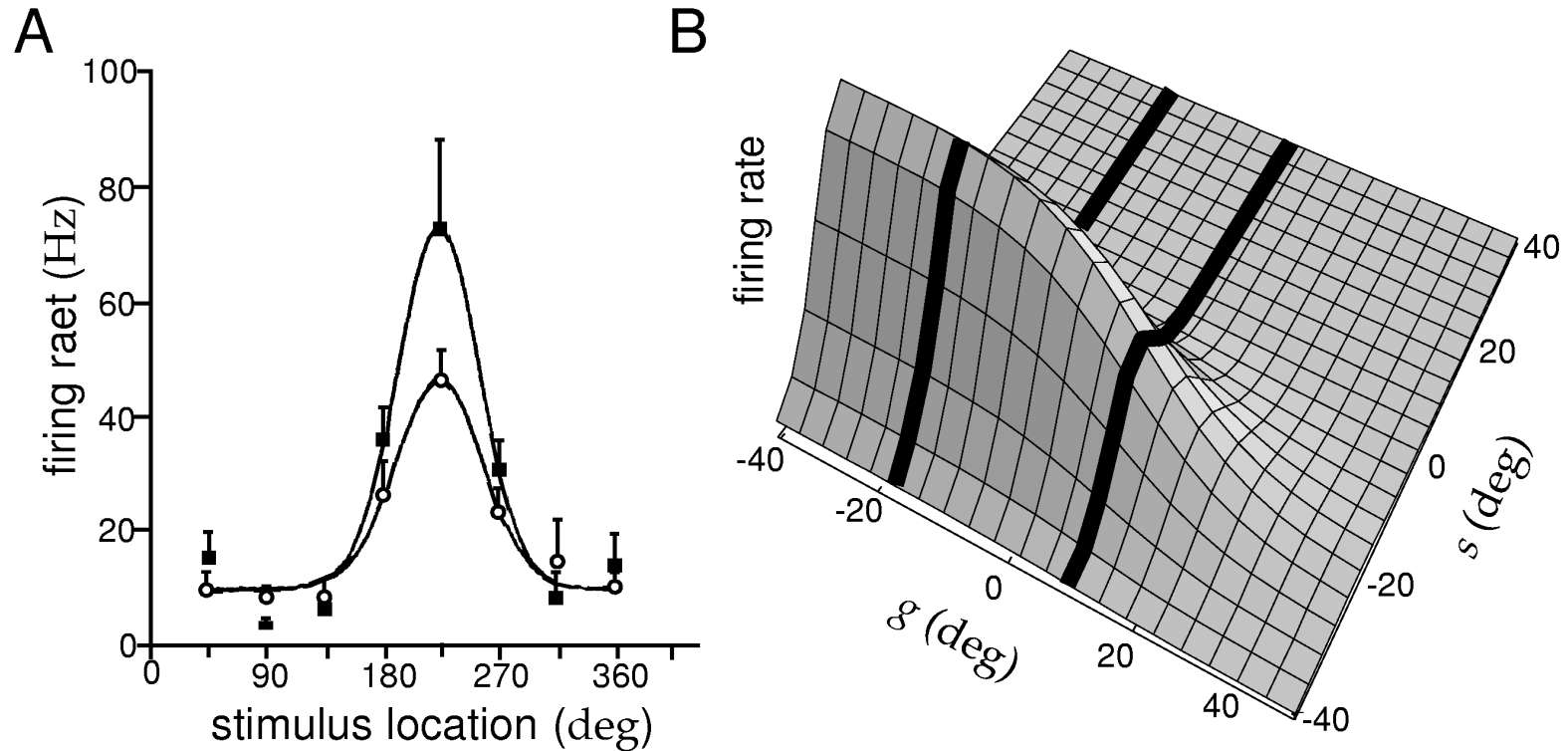
— Accounted by a feedforward network

Tuning curves of a visually responsive neuron in the premotor cortex of a monkey.



Gaze-dependent gain modulation

– of a retinotopic visual receptive field



Average firing-rate tuning curves of an **area 7a** neuron in **posterior parietal cortex**

A. Brotchie et al 1995

B. Pouget & Sejnowski 1995

Constructing the premotor neuron response

Average firing-rates of input neurons: $u = f_u(s - \xi, g - \gamma)$ with (ξ, γ) characterizes a neuron. The steady-state response of output neuron is

$$\begin{aligned} v_\infty &= F \left(\rho_\xi \rho_\gamma \int d\xi d\gamma w(\xi, \gamma) f_u(s - \xi, g - \gamma) \right) \\ &= F \left(\rho_\xi \rho_\gamma \int d\xi d\gamma w(\xi - g, \gamma + g) f_u(s + g - \xi, -\gamma) \right) \end{aligned}$$

following the shifts $\xi \rightarrow \xi - g$, $\gamma \rightarrow \gamma + g$ of the integrations. If $w(\xi - g, \gamma + g) = w(\xi, \gamma) = w(\xi + \gamma)$, v_∞ is a function of $s + g$ only.

Input $(s, g) \rightarrow$ Area 7a neurons $\rightarrow$ Motor cortex $(s + g)$

Continuous linear recurrent networks

Continuous extension of the linear system in Lecture 13:

$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + h(\theta) + \rho_\theta \int_{-\pi}^{\pi} d\theta' M(\theta - \theta') v(\theta')$$

- Since θ is an angle, all functions are periodic with period 2π
- With M depending only on $\theta - \theta'$, symmetry is imposed in translations or shifts of the angle variables.
- Even $M(\theta - \theta') = M(\theta' - \theta)$ function is analogous to a symmetric synaptic weight matrix.

Eigenfunctions for continuous recurrent network

Generalize eigenvectors $\mathbf{M} \cdot \mathbf{e}_\mu = \lambda_\mu \mathbf{e}_\mu$ to continuous space:

$$\rho_\theta \int_{-\pi}^{\pi} d\theta' M(\theta - \theta') e_\mu(\theta') = \lambda_\mu e_\mu(\theta)$$

The eigenfunctions are $1/\sqrt{2\pi\rho_\theta}$ for $\mu = 0$ as well as $\cos(\mu\theta)/\sqrt{\pi\rho_\theta}$ and $\sin(\mu\theta)/\sqrt{\pi\rho_\theta}$ for $\mu = 1, 2, \dots$. The eigenvalues can be calculated as

$$\lambda_\mu = \rho_\theta \int_{-\pi}^{\pi} d\theta' M(\theta') \cos(\mu\theta')$$

Steady state for constant $h(\theta)$:

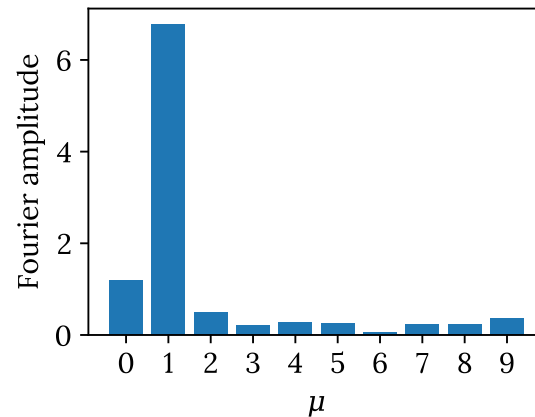
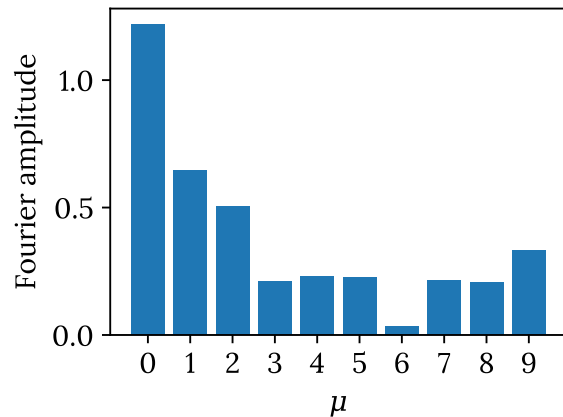
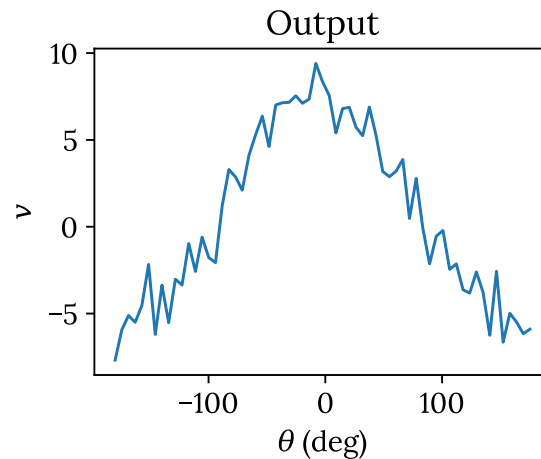
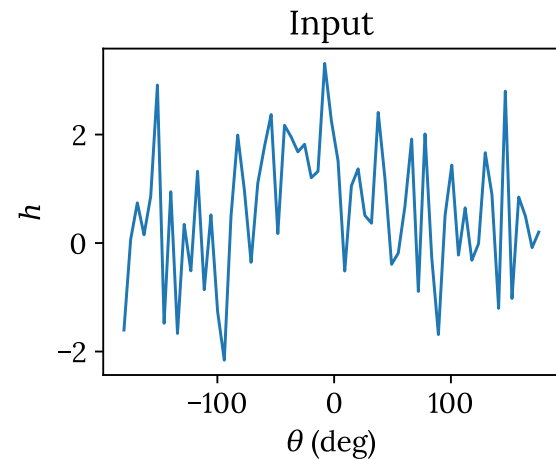
$$\begin{aligned} v_\infty(\theta) = & \frac{1}{1 - \lambda_0} \int_{-\pi}^{\pi} \frac{d\theta'}{2\pi} h(\theta') \\ & + \sum_{\mu=1}^{\infty} \frac{\cos(\mu\theta)}{1 - \lambda_\mu} \int_{-\pi}^{\pi} \frac{d\theta'}{2\pi} h(\theta') \cos(\mu\theta') \\ & + \sum_{\mu=1}^{\infty} \frac{\sin(\mu\theta)}{1 - \lambda_\mu} \int_{-\pi}^{\pi} \frac{d\theta'}{2\pi} h(\theta') \sin(\mu\theta') \end{aligned}$$

Cosine and sine
Fourier integrals

Example: selective amplification

Consider the connection

$$M(\theta - \theta') = \frac{\lambda_1}{\pi \rho_\theta} \cos(\theta - \theta')$$



with $\lambda_1 = 0.9$ and all other eigenvalues are zero, $\lambda_\mu = 0$ for $\mu \neq 1$.

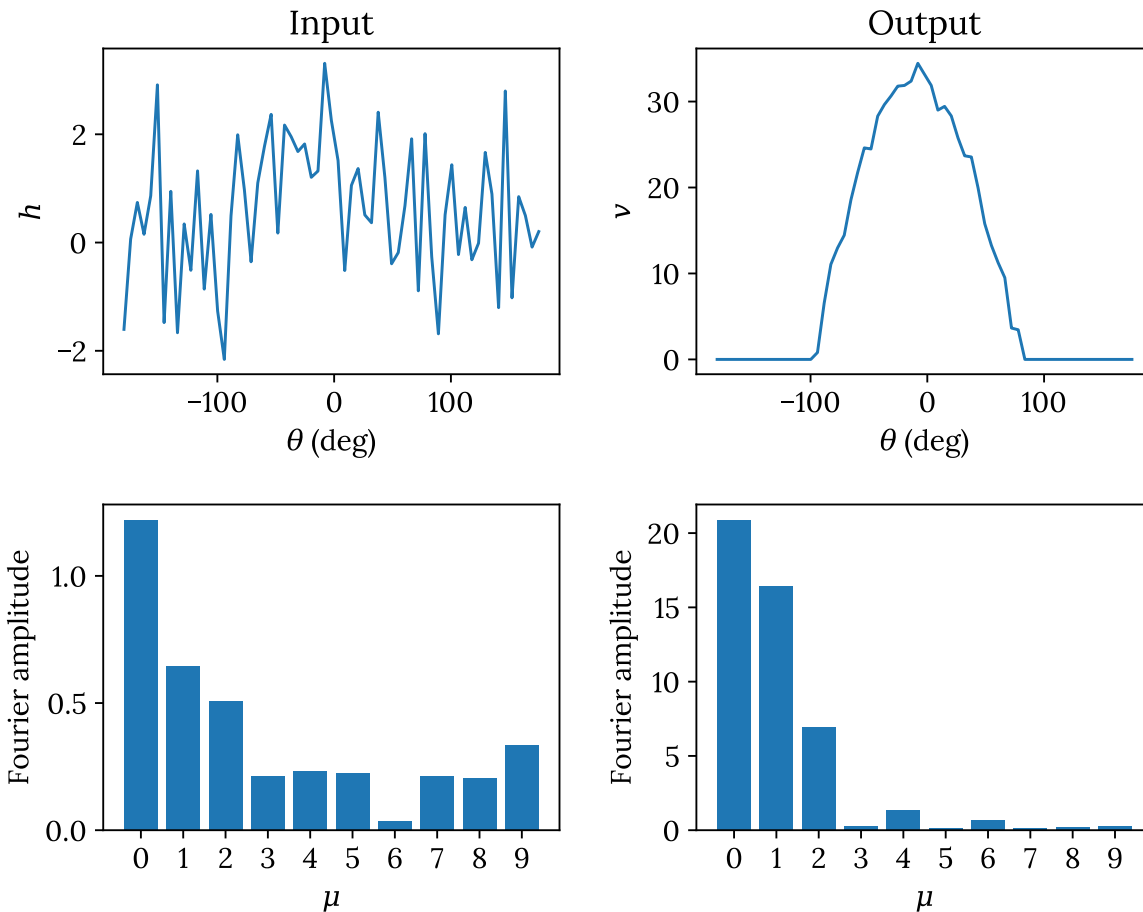
The $\mu = 1$ Fourier amplitudes of the input is amplified by a factor of 10 at the output.

Stability requires: $\lambda_\mu < 1$.

Nonlinear recurrent networks

With rectification

$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + \left[h(\theta) + \frac{\lambda_1}{\pi} \int_{-\pi}^{\pi} d\theta' \cos(\theta - \theta') v(\theta') \right]_+$$



with $\lambda_1 = 1.9$ and $\lambda_\mu = 0$ for $\mu \neq 1$.

Dynamics can still diverge but not determined by > 1 eigenvalues.

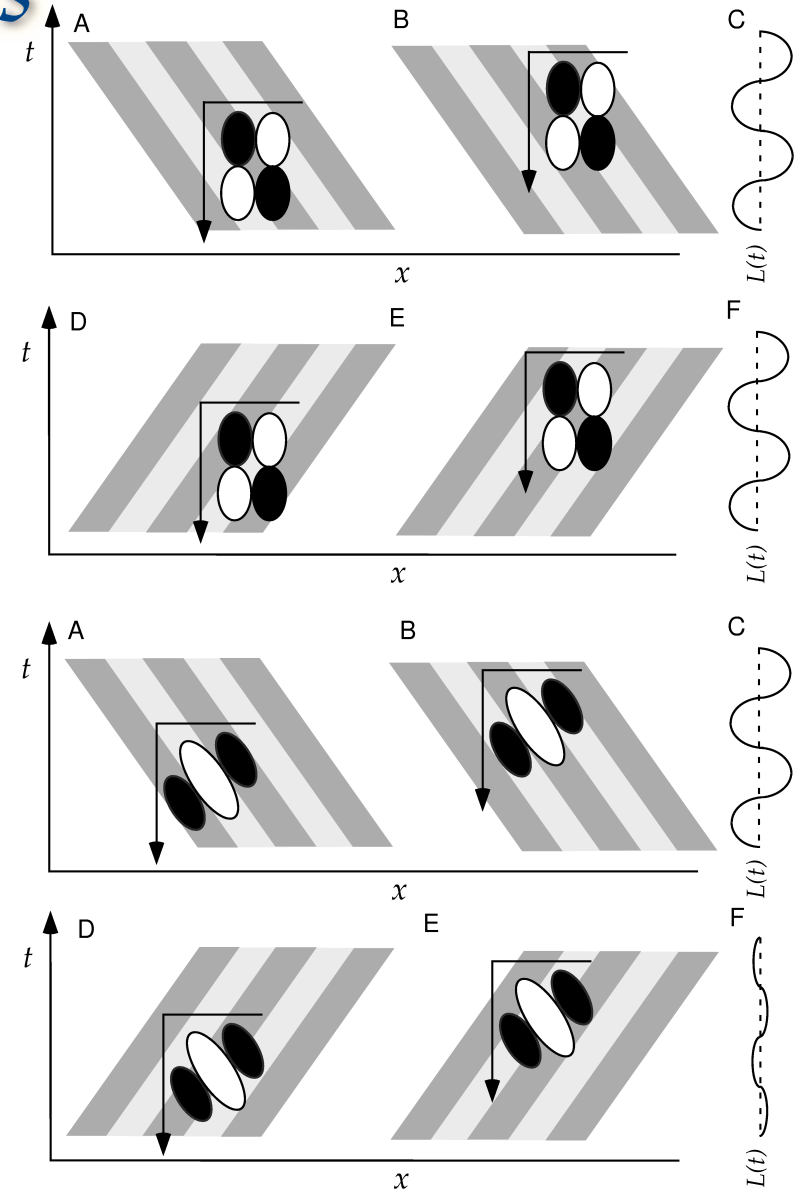
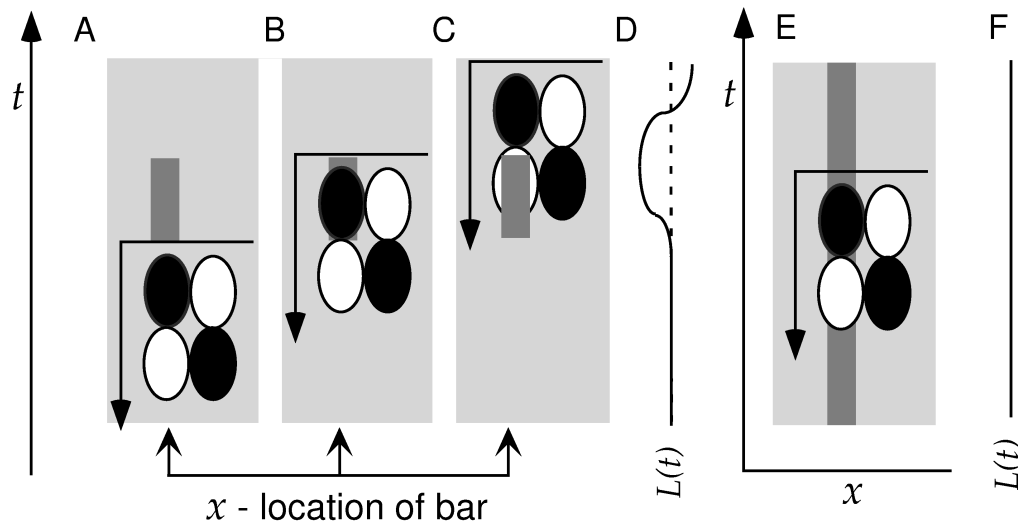
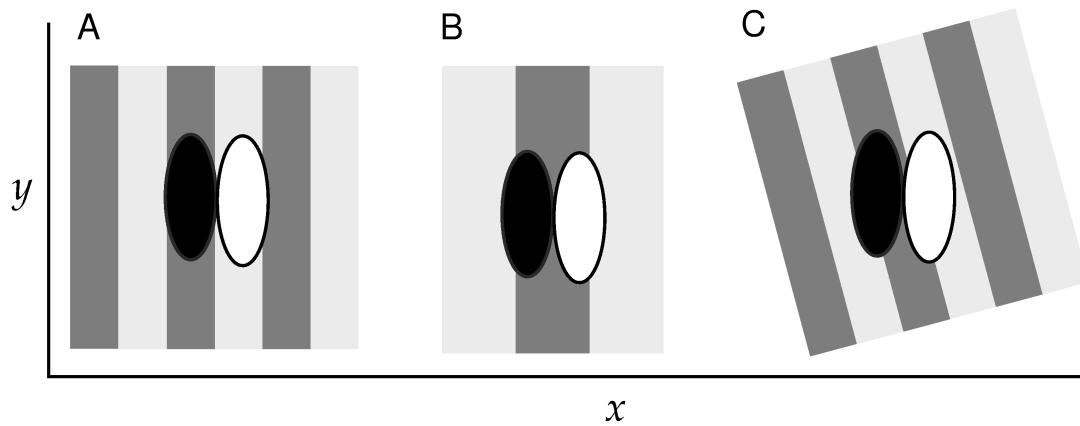
More ($\nu = 0, 1, 2$) Fourier components are amplified.

Simple cells

1. They have distinct excitatory and inhibitory regions.
2. These regions follow the **summation property**.
3. These regions have mutual antagonism — excitatory and inhibitory regions balance themselves out in diffuse lighting.
4. It is possible to predict responses of moving stimuli given **the map** of excitatory and inhibitory regions.

[wikipedia.org](https://en.wikipedia.org/wiki/Simple_cell)

Receptive fields of simple cells



Dayan & Abbott (2007)

Static nonlinearities

Simple cells with contrast saturation

Rate estimate, $r_{\text{est}}(t) = r_0 + F(L(t))$, Simplest: $F = G[L]_+$ where G is a fitting constant and L is the linear response of the neuron. With contrast saturation:

$$F(L) = \frac{G[L]_+^2}{A_{1/2}^2 + G[L]_+^2}$$

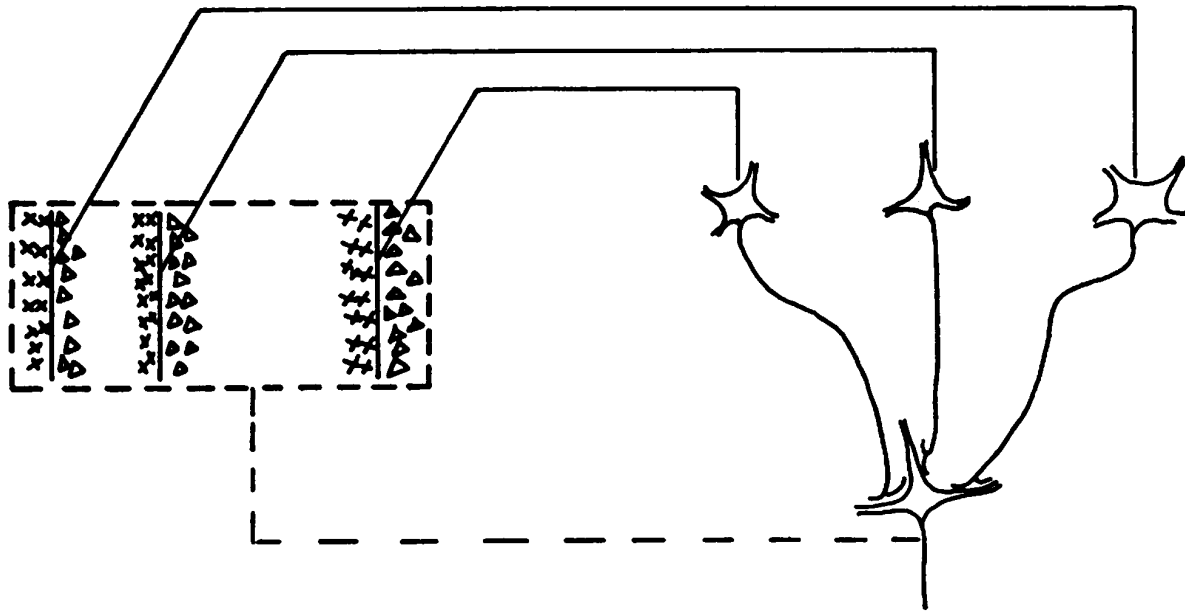
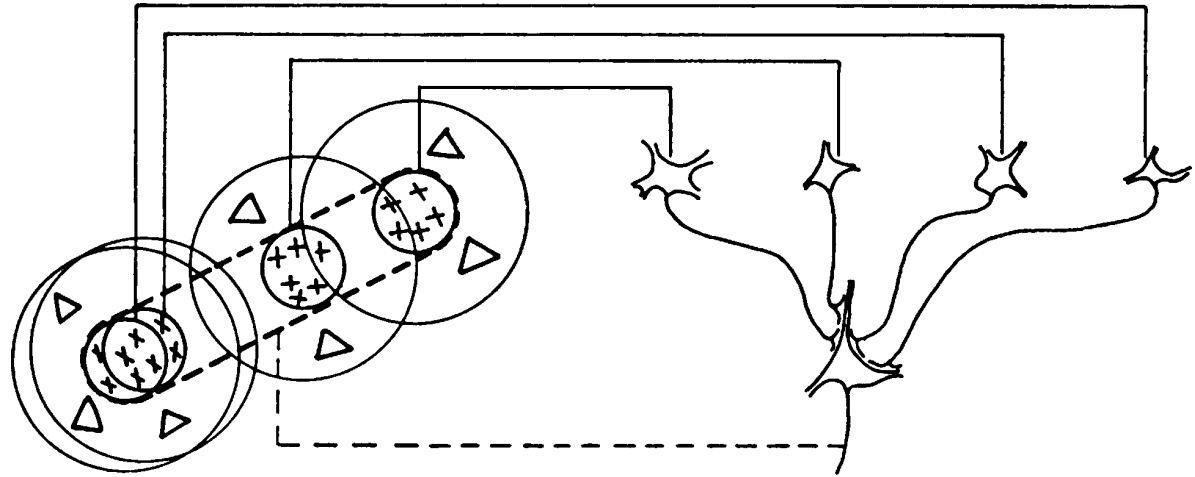
Complex cells

Behaviors are generally not explained by linear mechanism. For example, spatial-phase invariance, frequency doubling, ...

Can be selective for particular types of images independent of exact spatial positions within receptive fields → early stage of position invariance object recognition

Hubel–Wiesel model

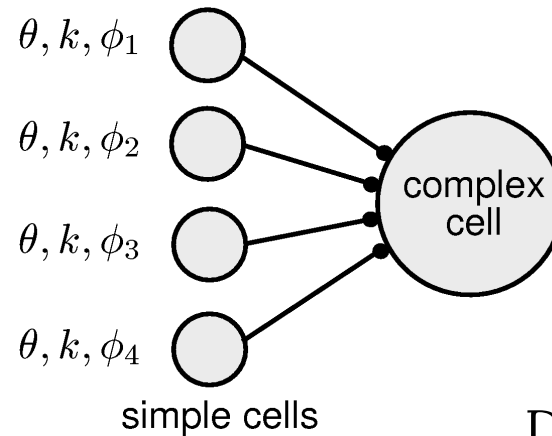
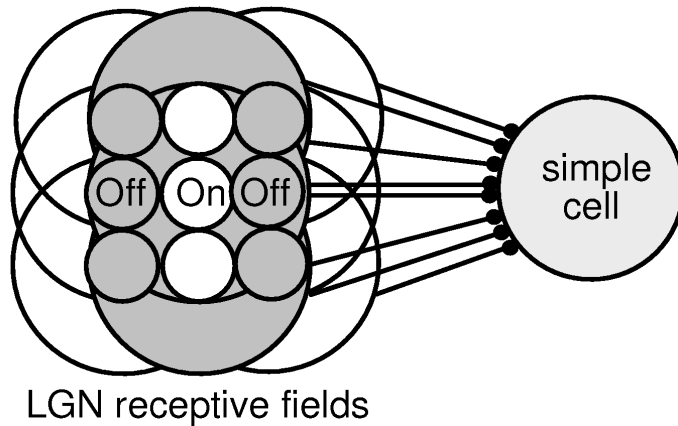
“Possible scheme for explaining ... **simple** receptive fields”



“Possible scheme for explaining ... **complex** receptive fields”

Hubel & Wiesel (1962)

Hierarchical organization & nonlinearity



Dayan & Abbott (2007)

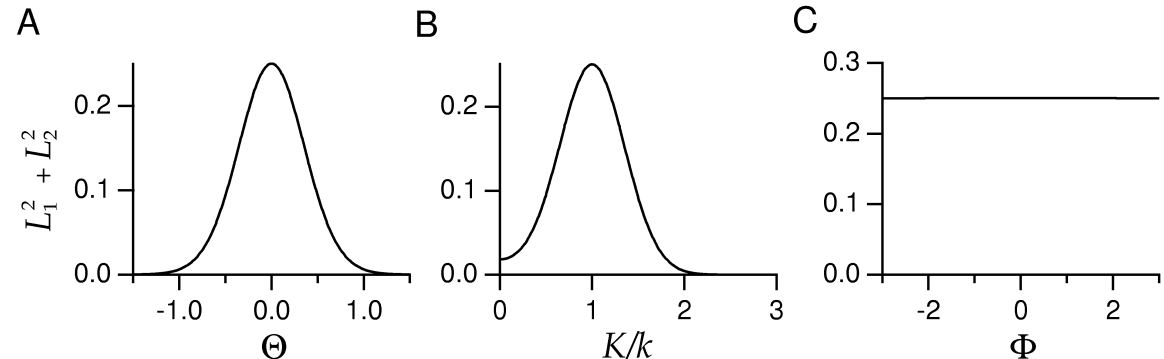
Simpler nonlinearity:

$$L_1 = AB(\omega, K) \cos(\phi - \Phi) \cos(\omega t - \delta).$$

$$L_2 = AB(\omega, K) \sin(\phi - \Phi) \cos(\omega t - \delta)$$

Sum of squares response:

$$r(t) = r_0 + G(L_1^2 + L_2^2).$$



A Recurrent Model of Simple Cells

$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + \left[h(\theta) + \int_{-\pi/2}^{\pi/2} \frac{d\theta'}{\pi} (-\lambda_0 + \lambda_1 \cos(2(\theta - \theta'))) v(\theta') \right]_+$$

Input $h(\theta)$: orientation tuned feedforward from ON-center and OFF-center LGN cells.

$$h(\theta) = Ac(1 - \epsilon + \epsilon \cos(2\theta))$$

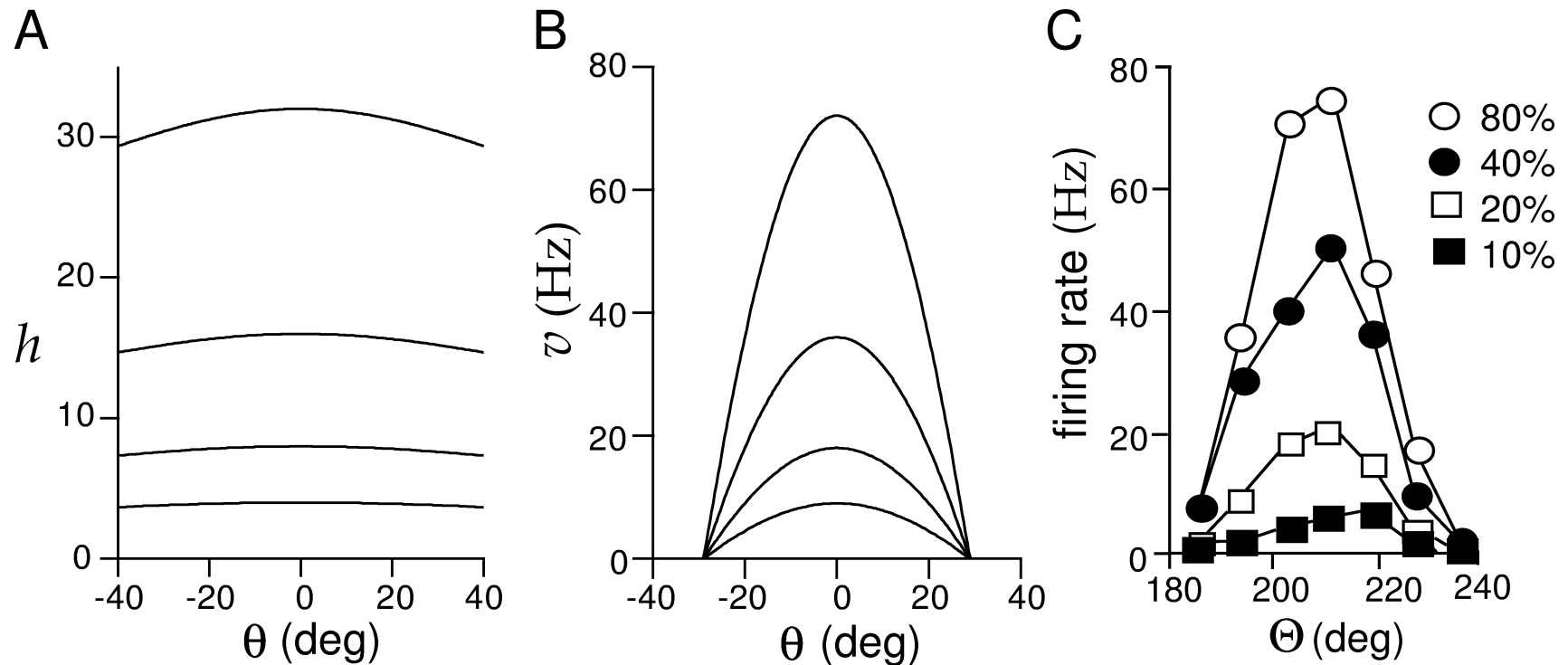
A : amplitude

c : image contrast

ϵ : strength of modulation

Because of the translational invariance considered, $v(\theta)$ can be interpreted as the tuning curve.

Contrast effect on orientation tuning

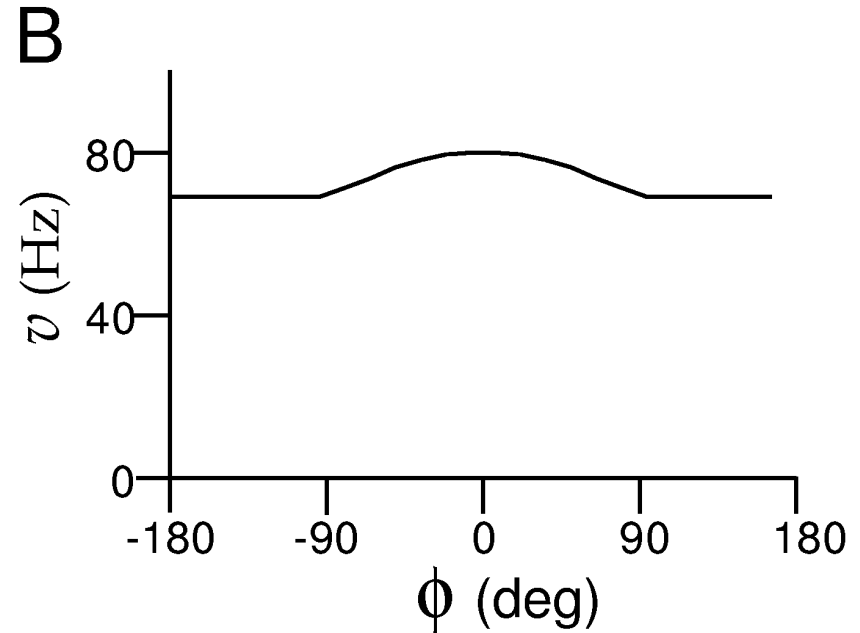
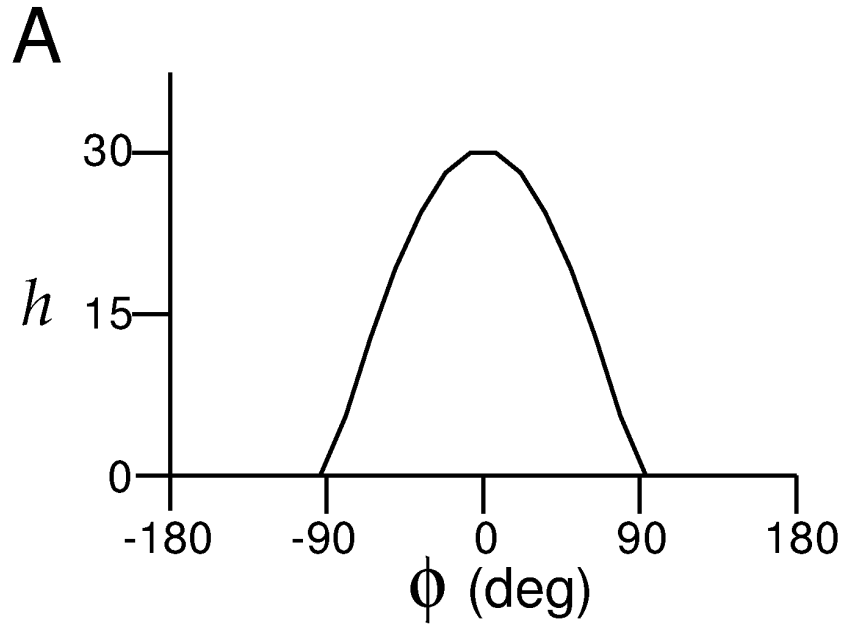


Parameters: $\lambda_0 = 7.3$, $\lambda_1 = 11$, $A = 40$ Hz, and $\epsilon = 0.1$.

C) Experimental tuning curves Sompolinsky and Shapley 1997

Recurrent model for complex cell

Chance et al (1999)



$$M(\phi - \phi') = \lambda_1 / (2\pi\rho_\phi)$$

$$\tau_r \frac{dv(\phi)}{dt} = -v(\phi) + \left[h(\phi) + \frac{\lambda_1}{2\pi} \int_{-\pi}^{\pi} d\phi' v(\phi') \right]_+$$

Implementation in Python

2D matrix of $\cos(\theta - \theta')$:

```
m = np.cos(np.subtract.outer(th, th))  
m = np.cos(th[:,None]-th[None,:])
```

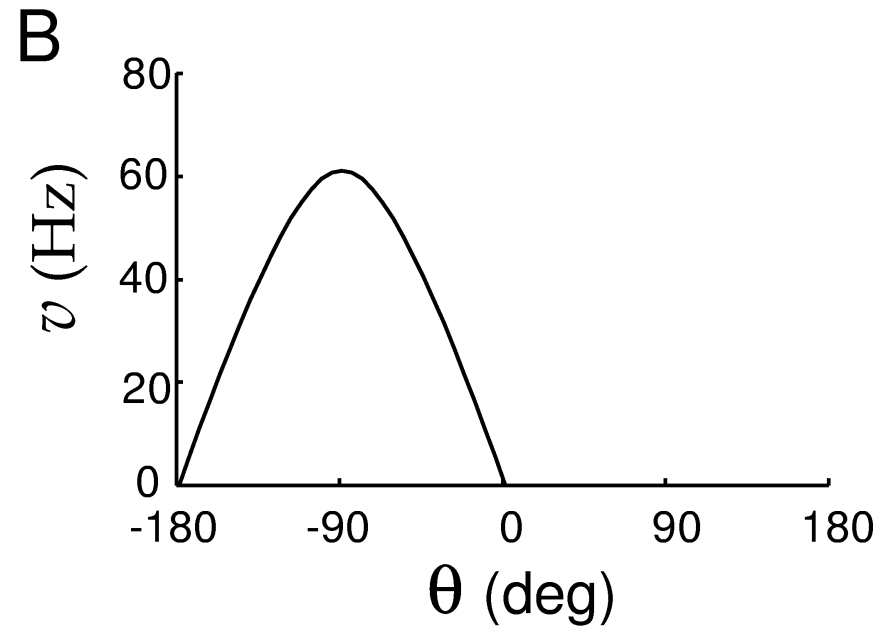
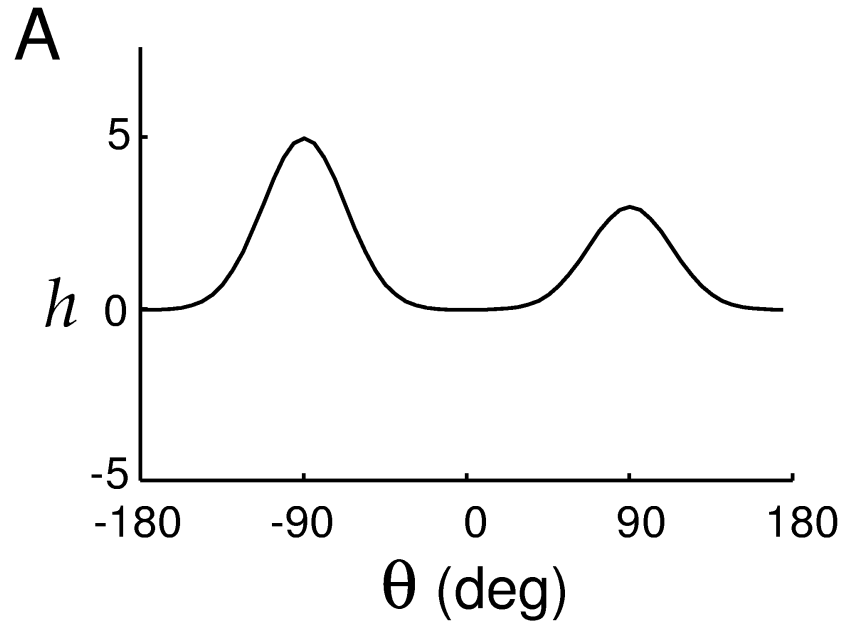
Rectified linear unit (ReLU):

```
x = (np.abs(x)+x)/2  
x[x<0] = 0
```

Integration becomes inner product between matrix and array:

```
dth*np.dot(m,v)  
dth*m@v
```

Winner takes all



Gain modulation

